DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS*****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

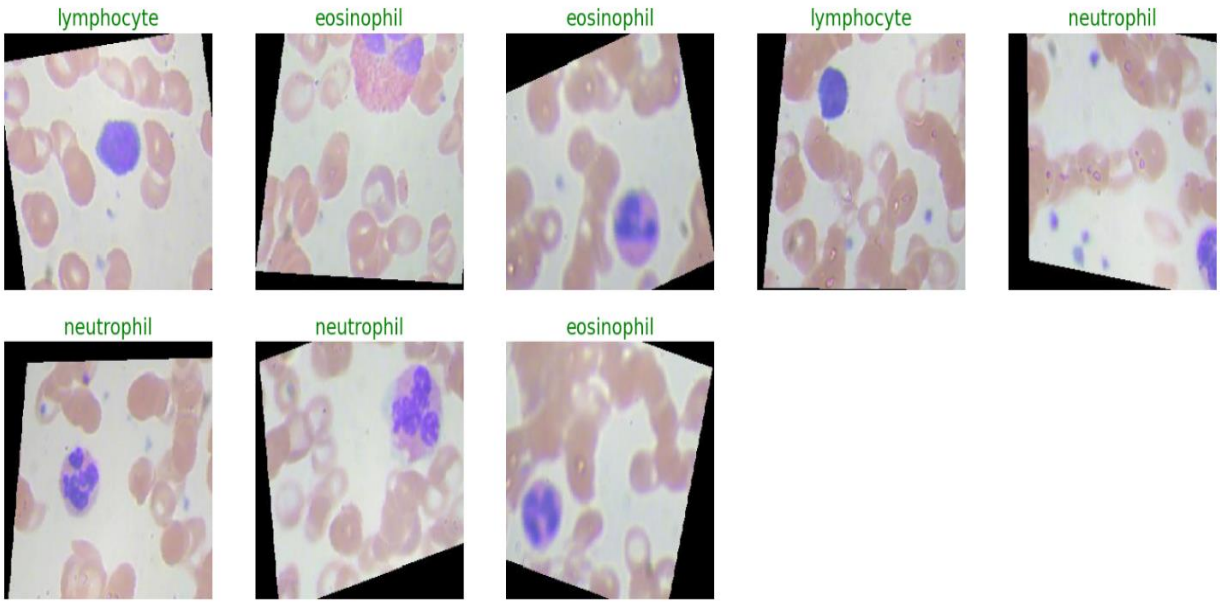
6.3 DATA PRE-PROCESSING :

DATA VISUALIZATION :

The provided Python code imports necessary libraries and modules for image manipulation. It selects a random image file from a specified folder path. Then, it displays the randomly selected image using IPython's Image module. This code is useful for showcasing random images from a directory for various purposes like data exploration or testing image processing algorithms.

```
import matplotlib.pyplot as plt

import numpy as np

def show_knee_images(image_gen):

    test_dict = test.class_indices

    classes = list(test_dict.keys())

    images, labels = next(image_gen)

    plt.figure(figsize=(20,20))

    length = len(labels)

    if length < 25:

        r = length

    else:

        r = 25

    for i in range(r):

        plt.subplot(5,5,i+1)

        image = (images[i] + 1) / 2

        plt.imshow(image)

        index = np.argmax(labels[i])
```

```
    class_name = classes[index]
    plt.title(class_name, color="green", fontsize=16)
    plt.axis('off')
  plt.show()
show_knee_images(train)
```



In the above code, I used class ace of diamond for prediction, This code randomly selects an image file from a specified folder (folder_path) containing JPEG, PNG, or JPEG files, and then displays the selected image using IPython's display function. It utilizes Python's OS and random modules for file manipulation and random selection, respectively. And It has predicted correctly as ace of diamond.